Viral Infections

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38.1 Herpes Viruses

38.1.1 Cytomegalovirus (CMV)

38.1.1.1 Clinical Symptoms

CMV can cause symptoms from almost any organ as well as nonspecific symptoms such as fever, malaise, and bone marrow suppression in stem cell transplant patients. However, the most important clinical entities in allo-HCT patients are pneumonia and gastroenteritis.

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The likelihood for symptomatic infection is much higher after allo-HCT compared to auto-HCT. Being CMV seropositive (CMV (+)) has traditionally been associated with decreased OS after allo-HCT. In addition, CMV replication is also associated with decreased OS and increased NRM. The situation has changed with the introduction of effective antiviral prophylaxis (see below). In patients undergoing MAC allo-HCT, the use of a CMV (−) donor to a CMV (+) patient has been associated with an increased risk for NRM and decreased OS.

38.1.1.2 Diagnostics

CMV antibody status should be determined pre-transplant in all patients undergoing HCT and in allogeneic stem cell donors.

Allo-HCT patients should be monitored weekly for CMV at least during the first 3 months after HCT. Patients with GVHD and those with documented CMV replication should be monitored longer. There is no need to routinely monitor patients after autologous HCT.

Today, the most commonly used technique is qPCR. Recently tests detecting CMV-specific T-cells have become available, but further evaluation of these tests' usefulness in routine care is necessary.

To diagnose CMV disease, it is important to combine symptoms and signs with documentation of the presence of CMV in affected tissue. An exception is CMV retinitis where ophthalmologic findings are characteristic although to detect CMV

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in vitreous fluid is helpful. Established techniques for the detection of CMV in tissue are histopathology, immunohistochemistry, and DNA hybridization. High levels of CMV DNA in BAL are associated with CMV pneumonia, while its absence almost excludes CMV pneumonia. PCR in CSF supports the diagnosis of CMV encephalitis. For other end-organ diseases, qPCR needs additional study.

38.1.1.3 Prophylaxis

If possible, a CMV seronegative donor should be chosen for a CMV seronegative patient. CMV safe blood products should be used.

Letermovir given for 3 months after HCT reduces the risk for clinically significant CMV infection (need for preemptive antiviral therapy and/or CMV disease) and also all-cause mortality in CMV (+) patients (Marty et al. 2017). Longer duration of prophylaxis can be given in high-risk patients. It is important to monitor patients after stopping letermovir prophylaxis since reactivations are then common. Ganciclovir can reduce the risk for CMV disease but is associated with significant toxicity. The data regarding prophylactic Ig is conflicting, and its use is not recommended.

38.1.1.4 Treatment

Ganciclovir, valganciclovir, and foscarnet have all been shown to be effective to prevent the development of CMV disease in allo-HCT recipients when given pre-emptively after detection of CMV in blood. Their efficacy is similar, so the choice should be based on the risk for side effects and practical aspects.

It is not possible to give a recommendation on what CMV DNA level preemptive therapy should be initiated since this depends on patient factors, the material used for monitoring (plasma/whole blood), and the performance of the assay used (Ljungman et al. 2019).

Therapy is usually given for at least 2 weeks, but longer therapy courses might be needed. Repeated reactivations are not uncommon, and either of the drug mentioned above can be used for retreatment. Maribavir was in a randomized controlled trial shown to be more effective than other alternatives against resistant or refractory CMV infection and associated with less toxicities (Avery et al. 2022).

Ganciclovir (valganciclovir) and foscarnet have been the most used drugs for CMV disease. The addition of high-dose Ig for treatment of CMV pneumonia has been commonly used, but the data supporting this combination is limited. There is no data supporting the addition of Ig to antiviral treatment for other types of CMV disease. Maribavir is an alternative for resistant or refractory disease. Cidofovir has been used when other antiviral therapies failed (Ljungman et al. 2019).

The duration of therapy has to be decided on a case-by-case basis, but normally longer therapy is needed compared to preemptive therapy (6–8 weeks).

38.1.1.5 Cellular Immunotherapy

Transfer of CMV-specific T-cells has in several non-controlled studies been used to manage resistant/refractory CMV infection and disease following allo-HCT. The T-cells were initially derived not only from the HSC donor but also either uni- or multispecifc T cells from a third-party donors have been used. The efficacy in patients receiving high-dose (≥2 mg/kg) corticosteroids is likely to be low.

38.1.2 HHV-6 A and B

38.1.2.1 Clinical Symptoms

HHV-6B primary infection is the main cause of *exanthema subitum* in young children. It has also been associated with febrile seizures. Almost all children are infected by the age of 2 years. HHV-6A primary infection has so far not been associated with specific symptoms.

HHV-6B is the main cause of viral encephalitis after allo-HCT, but HHV-6A has also been documented. Patients undergoing CBT are at an increased risk. Other symptoms suggested to be associated with HHV-6 are bone marrow suppression, pneumonia, and acute GVHD.

38.1.2.2 Diagnostics

Serology is not useful. HHV-6 DNA can be analyzed in blood by qPCR. However, the usefulness of monitoring is not established. HHV-6 can be integrated in germline. These individuals are strongly

positive in qPCR, but this is not a proof of viral replication.

MRI is recommended for diagnosis of HHV-6 encephalitis. The typical finding is of limbic encephalitis, but other patterns are also seen. HHV-6 DNA is usually positive in the CSF in patients with encephalitis.

38.1.2.3 Prophylaxis

Foscarnet has been used, but its usefulness is not established.

38.1.2.4 Treatment

Either ganciclovir or foscarnet can be used for treatment of HHV-6 encephalitis. There is no established treatment for HHV-6 infection or patients with other suspected HHV-6-associated complications. Cellular immunotherapy was only performed in a few patients.

38.1.3 HHV-7

38.1.3.1 Clinical Symptoms

HHV-7 primary infection is very common in young children occasionally causing *exanthema subitum (roseola)* and rarely status epilepticus with fever. HHV-7 detection after HCT is infrequent, with rare cases in which HHV-7 has been associated with CNS disease (encephalitis, myelitis).

38.1.3.2 Diagnostics

HHV-7 DNA by qPCR. HHV-7 might be a cofactor of CMV reactivation.

38.1.3.3 Prophylaxis

Not used.

38.1.3.4 Treatment

Infection by HHV-7 does not require specific treatment.

38.1.4 HHV-8

38.1.4.1 Clinical Symptoms

HHV-8 (KSHV, Kaposi's sarcoma-associated herpesvirus) is the cause of Kaposi's sarcoma (KS), primary effusion lymphoma, and multicentric Castleman's disease. KS is very rare after HCT. Fever and marrow aplasia with plasmacytosis can occur. Skin involvement is the dominant clinical presentation in adults, while pediatric cases can have visceral involvement.

38.1.4.2 Diagnostics

Detection of HHV-8 DNA by qPCR. KS can be clinically defined on the basis of characteristic skin lesions or histopathologically defined in a malignant tumor.

38.1.4.3 Prophylaxis

Not recommended.

38.1.4.4 Treatment

In disease limited to the skin only, surgical excision or electrochemotherapy is the most preferable approach. For visceral or disseminated disease, possible options include the use of interferon alpha or chemotherapy. The use of antiviral treatment is considered without benefit. Imatinib showed promising results in HIV-related KS patients.

38.1.5 EBV

38.1.5.1 Clinical Symptoms

Syndromes caused by primary EBV infection include infectious mononucleosis, chronic active EBV infection, and X-linked lymphoproliferative syndrome.

In HCT patients, EBV can cause life-threatening complication: posttransplant lymphoproliferative disorder (PTLD) or end-organ diseases such as encephalitis/myelitis, pneumonia, or hepatitis. Details on EBV-PTLD are presented in Chap. 45.

Donor EBV seropositivity also contributes to the risk of cGVHD in patients with acute leukemia.

38.1.5.2 Diagnostics

All allo-HCT patients and donors should be tested for EBV Ab before HCT.

38.1.5.3 Prophylaxis

Since EBV sero-mismatch is a risk factor for PTLD, the selection of an EBV-matched donor,

if possible, might be beneficial. As EBV-PTLD after HCT is usually of donor origin and EBV might be transmitted with the graft, the risk of EBV-PTLD is higher when the donor is seropositive. Anti-CD20 antibodies have been used to prevent EBV reactivations in high-risk patients.

38.1.5.4 Treatment

Most EBV reactivations are subclinical and require no therapy. Antiviral therapy is not effective. Pre-emptive therapy with anti-CD-20 antibodies is one option to decrease the risk for EBV-PTLD. Treatment of EBV-PTLD is discussed in Chap. 45.

38.1.6 Herpes Simplex Virus (HSV)

38.1.6.1 Clinical Symptoms

HSV reactivations can be caused by either type 1 or 2 and is usually associated with localized mucocutaneous disease in the orofacial region (85–90%) and less frequently in the esophageal and genital area. Uncommon manifestations are pneumonia, hepatitis, meningitis (HSV-2), and encephalitis (HSV-1).

38.1.6.2 Diagnostics

All patients should be tested for HSV antibodies before HCT. The diagnosis of mucocutaneous HSV disease is suspected on clinical grounds, and the diagnosis is usually verified by PCR. PCR in CSF is the technique of choice for the diagnosis of HSV meningitis and encephalitis.

38.1.6.3 Prophylaxis

Primary HSV infection in HCT patients is unusual, and antiviral drug prophylaxis is thus not recommended in HSV-seronegative patients (but might be needed against VZV; see below). HSV-seropositive patients undergoing allo-HCT should receive antiviral drug prophylaxis. IV acyclovir 250 mg/m^2 or 5 mg/kg q12h, oral acyclovir 3 × 200 to 2 × 800 mg/day, oral valaciclovir 2 × 500 mg/day, or famciclovir 2 × 500 mg/day can be used.

The duration of prophylaxis depends on if prophylaxis against VZV is also indicated (see below) but should be given for at least 4 weeks after HCT in VZV-seronegative patients.

38.1.6.4 Treatment

IV acyclovir 250 mg/m^2 or 5 mg/kg q8h for 7–10 days is the therapy of choice for severe mucocutaneous or visceral HSV disease.

Oral acyclovir, from 5 × 200 to 5 × 400 mg/day, valaciclovir 2 × 500 mg/day, or famciclovir 2 × 500 mg/day for 10 days are considered as alternatives for less serious manifestations of HSV disease.

For HSV pneumonia or HSV meningitis and encephalitis, IV acyclovir 500 mg/m^2 or 10 mg/kg q8h for at least 14–21 days is recommended.

HSV resistance occurs in approximately 5–15% of patients and is mediated through mutation in the HSV thymidine kinase. Foscarnet or cidofovir is second-line therapies.

38.1.7 Varicella-Zoster Virus (VZV)

38.1.7.1 Clinical Symptoms

Primary infection (varicella) rarely occurs after HCT, but it might cause severe visceral disease.

Reactivations are common unless long-term antiviral prophylaxis and usually present as herpes zoster (shingles) and can be complicated by prolonged neuralgia. However, severe symptoms including disseminated infection similar to varicella, visceral disease presenting as severe abdominal pain or acute hepatitis, and rarely encephalitis, retinal necrosis, or pneumonitis can occur.

38.1.7.2 Diagnostics

Patients should be tested for VZV antibodies before HCT. The rash in clinical varicella or zoster is usually characteristic. However, in some cases, disseminated HSV can have a similar appearance. PCR on vesicular material for VZV and HSV can differentiate.

Visceral VZV disease can occur without rash and then PCR on blood is diagnostic.

38.1.7.3 Prophylaxis

VZV-seropositive patients should be given antiviral prophylaxis for at least 12 months or up to the end of IS therapy.

Prophylaxis can be given with acyclovir (2 × 800 mg; in children 2 × 20 mg/kg) or valacyclovir (2 × 500 mg).

Vaccination with recombinant VZV vaccine can be given to prevent late reactivations after discontinuation of antiviral prophylaxis but data is still limited in allo-HCT recipients.

In seronegative patients exposed to VZV, post-exposure prophylaxis with acyclovir or valacyclovir is recommended. Prophylaxis should be started as soon as possible and continued until 21 days after exposition.

38.1.7.4 Treatment

First-line therapy for varicella, disseminated zoster, and visceral disease is acyclovir 3 × 500 mg/m^2/day IV.

For localized or limited infections, oral valaciclovir (3 × 1000 mg), acyclovir (5 × 800 mg; in children 4 × 20 mg/kg), or famciclovir (3 × 500 mg) can be given until the lesions have crusted over (usually 7–10 days).

In case of resistance to acyclovir (rare), second-line therapies are foscarnet (60 mg/kg q12h) or cidofovir (5 mg/kg weekly, together with probenecid and hydration).

VZIg is not recommended. Only case reports exist on cellular therapy for VZV infection.

38.2 Community-Acquired Respiratory Viruses (CARVs) (Excluding SARS-CoV-2)

38.2.1 Epidemiology

HCT recipients, in particular allo-HCT recipients, are likely to contract CARV infections (by rhinovirus/enterovirus, RSV, seasonal coronavirus, parainfluenza virus, influenza, metapneumovirus, bocavirus, and adenovirus) with similar seasonality as in the community (Fontana and Strasfeld 2019). Respiratory symptoms are mostly mild causing only URI (above the larynx) but LTD (below the larynx) occurs in 10–30% of cases, and some recipients can develop life-threatening symptoms. CARV LTD-related mortality is usually low (<5%), but it could be >30% in recipients with profound immunosuppressed status at the time of infection. Multiple CARVs co-infections are common, likely due to seasonal overlap of CARV circulation in the community along with the characteristic long viral shedding which occurs in up to 20%, in particular in those under corticosteroids and profound lymphopenia. Bacterial and fungal coinfections are not uncommon and should be ruled out during the diagnostic workup.

38.2.2 Diagnostics

A CARV surveillance program in HCT units could be of value in reducing direct and indirect effects by these infections (Piñana et al 2020). Currently, there is no firm evidence of poorer outcome in allo-HCT recipients infected with an specific CARV as compared to others. Thus, it seems reasonable that CARV screening would be based on syndromic multiplex PCR platforms.

38.2.3 Risk Factors (RFs)

RFs for progression to LTD and mortality included lymphopenia, neutropenia, corticosteroids use, active GvHD, timing of infection from transplant, older age, and coinfections. These immunosuppression conditions should be assessed at the time of CARV infections for treatment decision-making and/or close clinical monitoring.

38.2.4 Management and Prevention

38.2.4.1 Pretransplant

Routine pretransplant radiology is advisable before transplant in symptomatic transplant candidates with CARV to rule out the presence of LTD. Transplant should be delayed until resolution in symptomatic candidates with LTD.

If antiviral therapy is available, recipients with UTD should be treated to shorten clinical symptoms before transplant (i.e., influenza virus, RSV, and SARS-CoV-2).

For CARVs UTD, with the exception of rhinovirus and common coronavirus, it is reasonable to delay transplant until symptoms resolution.

However, for rhinovirus and common coronavirus UTD, a risk/benefit should be considered before delaying HCT. There is no clear clinical benefit in vaccinating HCT recipients before transplant.

38.2.4.2 Posttransplant

For most of the CARVs, there are no effective antiviral drugs and/or vaccines and management and prevention should be focused in supportive care, bacterial or fungal LTD coinfection therapy, and preventive transmission measures to limit outbreak situations.

38.2.4.3 Influenza

Prophylaxis

The most important prophylactic measure is yearly vaccination with inactivated influenza vaccine preferably given at 6 months after HCT, although it can be considered earlier in outbreak situations. A second dose of vaccine can be considered.

Treatment

Standard therapy is with neuraminidase inhibitors, mainly oseltamivir or zanamivir, as soon as possible during the course of the disease. It should be recognized that the normally recommended duration of 5 days often is too short since viral excretion might continue for a long time. Resistance to oseltamivir is not rare although variable with the strain circulating in that particular season.

38.2.4.4 RSV

Prophylaxis

Recently, positive results have come out from trials with RSV vaccines against RSV in adults. None is currently available. There is no indication for prophylaxis with anti-RSV monoclonal antibodies.

Treatment

Ribavirin either given as inhalation or systemically +/− iv Ig has been suggested to reduce the risk for progression of RSV UTI to LTD and possibly to reduce mortality in RSV pneumonia.

38.2.4.5 Human Parainfluenzavirus (hPIV) and Metapneumovirus (HMPV)

hPIV exists in four different types of which especially type 3 can produce higher LRTD and mortality. There is no effective prophylaxis. Ribavirin could be considered in high-risk patients although data regarding efficacy is weak.

Similarly, HMPV has comparable LRTD and mortality rates to influenza and RSV. There is no effective prophylaxis available. Ribavirin's effectiveness as therapy is still uncertain.

38.3 Severe Acute Respiratory Syndrome Coronavirus-2 (SARS-CoV-2) Infection and Coronavirus 2019 Disease (COVID-19)

38.3.1 Clinical Symptoms

COVID-19 is caused by SARS-CoV-2, which was responsible for the recent pandemic causing millions of deaths worldwide. It is spread not only by droplets but also by contact with infected fluids. The usual clinical presentation is a flu-like syndrome with fever, cough, and fatigue. Atypical symptoms are diarrhea, vomiting, confusion, loss of appetite, taste, and/or smell. Asymptomatic infections are common.

COVID-19 may progress to a severe or critical course with breathlessness, interstitial pneumonia with or without consolidations, need for oxygen support, and intensive care. Death due to respiratory insufficiency or multiorgan failure has been reported in 12–25% of HCT patients, although it became substantially lower after introduction of vaccination and the emergence of the Omicron variants (Ljungman 2023). In patients who recovered from COVID-19 from more than 4 weeks, the persistence of fatigue, dyspnea, cough, chest pain, sleep disturbance, and declined quality of life after more is defined as long-COVID (or post-acute COVID-19 syndrome).

38.3.2 Diagnostics

The reference diagnostic test is the search of viral RNA on nasopharyngeal specimen by RT-PCR. The use of rapid antigen test is an alternative to RT-PCR test although less sensitive. Testing of lower respiratory tract fluids is recommended only for differential diagnosis for other clinical etiologies or in case of suspicion of coinfections. Assessment of SARS-CoV-2 serum antibody titers is not useful for diagnostic purposes, while it can be used for epidemiological investigations of seroprevalence or to assess the duration of immunity induced by wild infection or vaccine. Despite both natural infection and vaccination elicit a T-cell response, the role T-cell assays in the routine uses is not defined yet.

38.3.3 General Management

Preventing infection is based on the control measures recommended for aerosol-droplet-contact transmission: hand hygiene, social and physical distancing, face masks, and ventilation of rooms. Patients hospitalized for the treatment of COVID-19 should be cared for out of HCT ward and to prevent hospital outbreaks, possibly in single rooms with negative or neutral positive air pressure; moreover health care workers must wear protective equipment such as gloves, gowns, face shield, FFP2 mask, and practice careful disinfection of hands (Cesaro et al 2022).

In the SARS-CoV-2 positive patient, the deferral of cellular therapy (HCT, CAR-T) until the achievement of clinical and virological negativity is recommended to reduce the risk for severe-critical COVID-19. The resumption of cellular therapy program in the asymptomatic patient with persistent shedding of the virus requires a case-by-case risk/benefit assessment (Cesaro et al 2022).

38.3.4 Prophylaxis

Vaccination with mRNA vaccines represents the main measure to reduce the risk of SARS-COV-2 infection and to prevent severe and critical form of COVID-19. The primary immunization schedule is of 3 vaccine doses, starting from 3 to 6 months after transplant, followed by a booster dose after 3–4 months from the primary vaccine schedule. Additional booster doses may be scheduled according to country recommendations and epidemiological data.

In patients where the vaccination is contraindicated or considered ineffective and are at high risk of progression to severe critical COVID-19, monoclonal antibodies against anti-Spike protein are recommended as pre-exposure or post-exposure prophylaxis, provided that they are active against the circulating variant (Table 38.1).

38.3.5 Therapy

The therapy varies according to the risk profile of the patient and the phase of infection: directed to prevent or contain the viral replication in the early phase; directed to reduce the inflammatory response in the advanced phase. Moreover, supportive intensive cares are fundamental in the cases of severe-critical COVID-19. Table 38.1 shows the specific interventions recommended by ECIL 9 conference (Cesaro et al 2022).

38.4 Adenovirus (ADV)

Human ADV is double-stranded DNA viruses with worldwide distribution. They are divided into seven species (A-G), comprising more than 100 different virus types. Infections occur throughout the year. Nearly all people have evidence of prior infection by 10 years of age. Transmission can occur through fomites, aerosolized droplets, fecal-oral spread, infected tissue, or blood.

Following primary infection, ADVs can persist in different tissues, particularly tonsillar and adenoidal T-lymphocyte, from where active infection may recur in the presence of immunosuppression. In contrast to the rest of community-acquired respiratory viruses, adenoviral

Table 38.1 Management of patients with COVID-19

Pre-exposure prophylaxis (patient at high risk of severe-critical COVID-19)	• Monoclonal antibodies against anti-Spike protein, as long as they are active against the circulating variant(s)
Post-exposure prophylaxis (patient at high risk of severe-critical COVID-19)	• Monoclonal antibodies against anti-spike protein, as long as they are active against the circulating variant(s)
Mild/moderate COVID-19, no O_2 requirement	Early treatment with: • Antivirals (nirmatrelvir/ritonavir or remdesivir or molnupiravir or • Monoclonal antibodies against anti-spike protein, as long as they are active against the circulating variant(s)
Moderate COVID with O_2 required/severe COVID-19	• Dexamethasone (in case of worsening add a second immunosuppressant: anti-IL-6 (tocilizumab, sarilumab), or JAK—inhibitor JAK—inhibitor (baricitinib, tofacitinib), or anti-IL1 (anakinra) • Remdesivir • Monoclonal antibodies against anti-Spike protein, as long as they are active against the circulating variant(s) or high titer of convalescent plasma if monoclonals not available
Critical COVID-19	• Dexamethasone (in case of worsening add a second immunosuppressant: anti-IL-6 (tocilizumab, sarilumab) • Remdesivir • Monoclonal antibodies against anti-pike protein, as long as they are active against the circulating variant(s)

infections can occur by exogenous acquisition or by reappearance of persistent endogenous virus.

38.4.1 Clinical Symptoms

Invasive ADV infections are more common in pediatric HCT patients (6%–42%) than in adults (3%–15%), but the clinical manifestations can be equally severe. The true incidence may be underestimated, particularly in adults, due to lack of routine monitoring.

The spectrum of ADV disease in HCT patients ranges from mild gastroenteric or respiratory symptoms to severe hemorrhagic enteritis, hemorrhagic cystitis, nephritis, hepatitis, pneumonia, encephalitis, myocarditis, and multiple organ involvement. ADV disease is most commonly diagnosed within 100 days of HCT.

Risk factors for ADV infection/disease include haploidentical or URD graft, CBT, TCD, GVHD III–IV, severe lymphopenia, and treatment with alemtuzumab or anti-thymocyte globulin. Studies in children showed that the onset of invasive ADV infection is almost invariably preceded by the appearance and expansion of the virus in the gastrointestinal tract. But in adults, the role of the gastrointestinal tract as an important site of ADV reactivation and expansion remains unclear (Hiwarkar et al 2018).

38.4.2 Diagnostics

ADV-DNA by qPCR. Monitoring with qPCR of ADV viremia in PB is recommended on at least a weekly basis for patients with at least one risk factor, starting immediately posttransplant until immune reconstitution (CD3 > 300/mm3). qPCR is also recommended in case of clinical suspicion of ADV infection/disease. The great majority of pediatric centers performs routine screening and preemptive approach, but in adult centers, the screening is normally done when risk factors are present (Hiwarkar et al 2018).

38.4.3 Prophylaxis

Nonpharmacological prophylaxis is mandatory: strict isolation and hygienic measures in patients shedding the virus are absolutely necessary to prevent horizontal transmission and nosocomial outbreaks. Prophylactic antiviral therapy is not recommended.

38.4.4 Treatment

There is no approved therapy for ADV. Tapering of immunosuppression should be considered, whenever possible.

Patients, especially children, with increasing viral load and at least one risk factor, should receive preemptive antiviral treatment with cidofovir 3–5 mg/kg/week for 2–3 weeks and, thereafter, every other week. Patients with probable or proven ADV disease should be treated with IV cidofovir (5 mg/kg weekly for at least three doses; thereafter, every other week), together with hyperhydration and oral probenecid. Ribavirin is not recommended for ADV, but if ADV species C is detected, the addition of ribavirin to cidofovir may be beneficial.

Donor-derived ADV-specific CTLs are an option for clinically non-responding patients. Intravenous brincidofovir is under development for therapy of adenovirus infection.

38.5 Polyomaviruses

The human polyomaviruses are an increasing family of virus now including 14 members, with a high seroprevalence in the general population for most of them (60–100%). In immunocompromised individuals, several polyomaviruses can cause severe disease.

The two classic human polyomaviruses BK (BKV) and JC (JCV) are the polyomaviruses with recognized clinical implications in HCT. WU and KI polyomaviruses, although frequently detected in the respiratory tract, have shown no association with respiratory disease. Merkel cell polyomavirus (MCPyV), discovered in 2008, is the only human oncovirus in the Polyomaviridae family (associated with the Merkel cell carcinoma, a rare aggressive cutaneous neuroendocrine carcinoma). Thereafter, nine new human polyomaviruses were discovered with three of them related to skin diseases (Bartley et al 2023): trichodysplasia spinulosa virus (TSPyV (the etiologic agent of Trichodysplasia spinulosa, also known as cyclosporine-induced folliculodystrofy, a rare, disfiguring skin disorder); human polyomavirus 6 and 7).

38.5.1 Polyoma JCV

38.5.1.1 Clinical Symptoms

Reactivation of the ubiquitous, neurotropic John Cunningham polyomavirus (JCV), under conditions of impaired cellular immunity may cause progressive multifocal leukoencephalopathy (PML); a rare, opportunistic, and severe disease of the CNS. Profound suppression in cellular immunity may constitute a primary PML risk factor. In HCT, the estimated prevalence is 35 per 100,000 person-years. In allogeneic-HCT patients, the time to the manifestation of symptoms ranged from one to 60 months (median: 8 months).

PML awareness increased following the introduction of several new immunomodulatory treatments including natalizumab, rituximab, efalizumab, infliximab, brentuximab, tacrolimus, and MMF. More recently, it has been associated with CAR T therapy with a deduced incidence of 0.9 cases per 1000 CAR T patients, similar to the incidence of PML after natalizumab.

38.5.1.2 Diagnostics

JCV-DNA by PCR in CSF. Brain biopsy with demonstration of JCV DNA or positive histological evidence of JCV antigens in the CNS is required for a definitive diagnosis of PML.

A definite PML diagnosis requires all the following three criteria: compatible clinical features, compatible neuroimaging findings, and a positive PCR result for JCV in cerebrospinal fluid (CSF). For probable PML diagnosis, clinical or imaging criteria are allowed to be omitted requiring a positive PCR result for JCV CSF. Possible PML diagnosis consists either of CSF PCR positive result alone or if negative, compatible clinical features with compatible neuroimaging findings.

38.5.1.3 Prophylaxis

Not used.

38.5.1.4 Treatment

No specific treatment is available. Infection control relies on restoration of the host's immune competence, which cannot be attained in the majority of cases. The application of G-CSF may

facilitate immune reconstitution and JCV clearance in the CSF. JCV-specific CTLs have been used with promising results in a few patients. There is also positive preliminary experience with immune checkpoint-blocking antibodies (nivolumab and pembrolizumab) in PML.

38.5.2 BKV

BKV (See Chap. 51: Hemorrhagic Cystitis and Renal Dysfunction).

38.6 Norovirus

38.6.1 Clinical Symptoms

Noroviruses are the most common cause of foodborne disease and acute nonbacterial gastroenteritis worldwide.

Its prevalence was 2% in adults and up to 22% among pediatric transplant recipients with diarrhea, requiring hospitalization in 55% and ICU admission in 27%. Recurrence rate was 29%.

Risk factors: second HCT, intestinal GVHD, children.

Norovirus can cause severe, prolonged disease complicated by enteritis, fever, recurrent hospitalizations for dehydration, chronic diarrhea, acute renal failure, weight loss, malnutrition, pneumatosis intestinalis, peritonitis, secondary bacteremia, and death.

38.6.2 Diagnostics

Viral RNA by RT-PCT in the stool.

The major clinical concern is that misdiagnosis of norovirus gastroenteritis as GVHD would lead to an inappropriate increase in immunosuppression. Even with a biopsy, the differentiation could be difficult as the characteristic histologic feature of GVHD, namely, crypt apoptosis, is also sometimes seen in norovirus infection, although it should be limited to the small intestine as opposed to gut GVHD, which usually affects both the small and large bowel.

38.6.3 Prophylaxis

Nonpharmacological prophylaxis is mandatory. Strict isolation and hygiene measures in patients shedding the virus are necessary to prevent horizontal transmission and nosocomial outbreaks.

38.6.4 Treatment

Symptomatic. Some reports indicate oral human immunoglobulin therapy. Specific therapies are not available. When feasible, immunosuppression should be decreased.

38.7 Flaviviruses

Flaviviruses are RNA-positive single-stranded viruses. These include dengue viruses, Tick Borne Encephalitis virus (TBE), Zika virus, and yellow fever. The diagnosis and management of these viruses are not well documented in the HCT (Muhsen et al 2023). These viruses are mainly transmitted by mosquitoes or ticks. Therefore, HCT patients and potential donors should avoid travelling to endemic areas when possible. Vaccines are available against TBE or yellow fever.

38.7.1 Zika Virus (ZIKV)

38.7.1.1 Clinical Symptoms

It is transmitted mainly by *Aedes aegypti* mosquitoes but also by sexual contact, maternal-fetal, or blood transfusion and possibly by other tissues or organs donated by infectious donors. Infection in the healthy population typically results in a mild, asymptomatic, or a self-limiting febrile illness lasting 4–7 days. Infection can be followed by neurological consequences including Guillain–Barre syndromes. The main problem worldwide is the development of microcephaly or other congenital neurological syndromes in children whose mothers have been infected during pregnancy.

38.7.1.2 Diagnostics

Direct detection of ZIKV-RNA in the initial period (first 7 days of symptoms) or serological testing.

38.7.1.3 Prevention

Blood, tissues, and cells should not be imported from areas of ZIKV transmission or should be tested negative for the presence of ZIKV. A donor diagnosed with ZIKV infection or who has just returned from an affected area should be deferred for at least 28 days after cessation of symptoms following the WHO recommendations. The deferral should be at least 3 months after sexual contact with person at risk.

38.7.1.4 Treatment

No specific prophylaxis or therapy is available.

38.7.2 Tick-Borne Encephalitis Virus (TBEV)

TBIEV is a zoonotic disease transmitted by the bite of infected ticks (*Ixodes Ricinus, I. persulcatus*) found in forests and in rare instances may be acquired by consumption of infected unpasteurized dairy products. Vertical transmission from an infected mother to the fetus is possible. Transmission by organ transplantation has been described (3 cases). It is endemic in many parts of Europe (with the highest incidence and increasing in the Nordic, Baltic, and Central European countries) and Asia. In Europe, most cases occur from May to November, with a peak in July.

38.7.2.1 Clinical Symptoms

TBEV causes Tickborne encephalitis (TBE). In healthy people, two-thirds of TBEV infections are asymptomatic. Symptomatic cases have two phases. The first viremic phase has non-specific symptoms (fever, fatigue, headache, myalgia, and nausea) and lasts approximately 5 days (2–10). It is followed by an asymptomatic period (7 days, range 1–33) that precedes the second "encephalitis phase," when the symptoms and signs of central nervous system involvement appear (meningitis, meningoencephalitis, myelitis, paralysis, and radiculitis). In Europe, 20–30% of patients experience the second phase with a mortality of 0.5–2%, but severe neurological sequelae are frequent and occur in up to 10% of patients. Risk factors for severe disease are age (>40 years) immunocompromised state. The course of the disease in HCT is unknown.

38.7.2.2 Diagnostics

The diagnosis of TBE is based on the detection of specific IgM antibodies in cerebrospinal fluid and/or serum or detection by PCR. Specific IgM antibodies can persist for up to 10 months in vaccinees or naturally infected individuals.

38.7.2.3 Prevention

TBEV can be prevented by avoiding tick bites (wearing protective clothing, using tick repellents) and avoiding consumption of unpasteurized dairy products in risk areas.

The most effective means of preventing TBE in endemic countries is by vaccination against TBEV (inactivated vaccine) particularly indicated in patients engaged in outdoor activities. There has been a clinical trial with the vaccine in HCT recipients starting at 9 months after transplantation, showing that it is safe although the response is lower compared to healthy individuals.

38.7.2.4 Treatment

Supportive care. No specific therapy is available.

38.8 West Nile Virus (WNV)

WNV infection is a mosquito-borne zoonosis, transmitted among birds via the bite of infected culex-species mosquitoes and incidentally to humans. The virus can also be transmitted by blood transfusion and organ donation, and even hematopoietic progenitors for HCT. Europe is endemo-epidemic and affects countries in southern, eastern, and western Europe. The incidence of WNV waxes and wanes. About 80% of WNV infections in humans are asymptomatic. West Nile fever (20%) is the most common clinical

presentation. The elderly and immunocompromised persons are at higher risk of developing West Nile neuroinvasive disease. Mortality in patients with neuroinvasive disease may reach up to 10%. The incidence of neuroinvasive disease in HCT is unknown. WNV should be included in the differential diagnosis of meningoencephalitis or lower extremity paralysis in HCT patients.

According to an EU-Directive, prospective blood donors should be deferred for 28 days after leaving a risk area for locally acquired WNV infection, unless the result of an individual nucleic acid test is negative. The same should be applied for a donor for HCT.

The diagnosis of WNV depends on a high index of suspicion and laboratory testing (serum and CSF WNV IgM and IgG antibodies and viral nucleic acid testing). For the diagnosis of WNV neuroinvasive disease, CSF should be studied.

No specific prophylaxis or treatment exists against the disease. The primary treatment of WNV is supportive care, although IVIG with high titles of WNV antibodies can be considered. Temporary reduction in immunosuppression should be considered.

38.9 Human T-Cell Lymphotropic Virus (HTLV)

HTLV-1 (human T-cell leukemia/lymphoma virus type 1) was the first oncogenic human discovered. It is present worldwide, with an estimated 5–10 million people infected. It has a high endemicity in Japan, the Caribbean region, areas of South America and tropical Africa and foci in the Middle East, Australia, and Melanesia. In Europe, the only country with an endemic HTLV-1 region is Romania. HTLV-1 is transmitted by mother-to-child, mainly linked to prolonged breast-feeding, sexual, and via transplantation of organs, tissues, and leucocyte-rich blood components.

HTLV-1 is the etiological agent of two diseases: adult T-cell leukemia/lymphoma (ATLL) and Tropical Spastic Paraparesis/HTLV-1-associated myelopathy.

The relevance to HCT comes from 2 sides. First, this virus can be transmitted by blood and hematopoietic progenitors. That affects the screening of donors. And second, for ATLL only, allogeneic HCT appears to be curative. Persons with HTLV-1 infection should be permanently deferred from donation of blood and blood components, although routine screening of blood donation is not recommended. Anti-HTLV-1 screening should be attempted in donors from geographical regions with a high prevalence of HTLV-1 infection or with sexual partners originating from those areas or where the donor's parents originate from those areas.

Key Points

- Epidemiology: Latent (especially CMV) and endemic (especially CARV including SARS-CoV-2 and ADV) viruses are important pathogens after HCT
- Diagnosis: Viral diagnostics after HCT require qPCR or multiplex PCR
- Prophylaxis and treatment: Prophylaxis (pharmacological or environmental) or preemptive treatment (if available) is necessary. All patients after HCT should undergo vaccinations according to current recommendations
- Outcome: Viral infections contribute to non-relapse mortality after HCT
- Age: new innovative drugs (letermovir) not yet approved for children

38.10 Viruses Covered in Other Chapters

BKV (See Chap. 51: Hemorrhagic Cystitis and Renal Dysfunction).

HIV (see Chap. 88: *Other T- and B-Aggressive Lymphomas and Lymphomas Associated with HIV*).

Hepatotropic Viruses (See Chap. 49: Hepatic Complications).

EBV (See Chap. 45: Posttransplant Lymphoproliferative Syndromes).

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